The Bounded-Trajectory Conditioning in Conjecture 3.18 Is Null

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17 August 2026

Status: candidate full negative resolution as written.

1 The conjecture

Jorgensen and Pearse define a resistance network to be a connected weighted graph with symmetric conductances $c_{zw} \geq 0$ and $c(z) := \sum_w c_{zw} < \infty$. Its random walk has transition probabilities $p(z,w) = c_{zw}/c(z)$. Their Conjecture 3.18 asks for the identity

$$v_x(y) = R^{\mathrm{F}}(o,x)\,\mathbb{P}_y[\tau_x < \tau_o \mid |\gamma| < \infty]\,, \tag{1}$$

where $|\gamma| < \infty$ means that the entire infinite trajectory is contained in some finite subnetwork [1, p. 22, Conj. 3.18].

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The error is discussed in Remark 3.19, just below. In the statement of Conjecture 3.18, we use the notation

$$|\gamma| < \infty \tag{3.25}$$

to denote the event that the walk is bounded, i.e., that the trajectory is contained in a finite subnetwork of G.

Conjecture 3.18. On an infinite resistance network, let v_x be the representative of an element of the energy kernel specified by $v_x(o) = 0$. Then for $x \neq o$, v_x is computed probabilistically by

$$v_x(y) = R^F(o,x)\mathbb{P}_y[\tau_x < \tau_o \,|\, |\gamma| < \infty], \tag{3.26}$$

that is, the walk is conditioned to lie entirely in some finite subnetwork as in (3.25).

"Proof." Fix x, y and suppose without loss of generality that $o, x, y \in G_1$. One can write (3.19) on G_k as $v_x^{(k)} = R_{G_k^F}(o,x)u_x^{(k)}$. In other words, $v_x^{(k)}$ is the unique solution to $\Delta v = \delta_x - \delta_o$ on the finite subnetwork G_k^F. Since $R^F(x,y) = \lim_{k\to\infty} R_{G_k^F}(x,y)$ by (2.10), it only remains to check the limit of $u_x^{(k)}$. Using a superscript to indicate the network in which the random walk travels, we have

$$\lim_{k\to\infty} u_x^{(k)}(y) = \lim_{k\to\infty} \mathbb{P}_y^{G_k^F}[\tau_x < \tau_o] = \lim_{k\to\infty} \mathbb{P}_y^{G}[\tau_x < \tau_o \,|\, \gamma \subseteq G_k^F]. \tag{3.27}$$

Here again, the notation $[\gamma \subseteq H]$ denotes the event that the random walk never leaves the subnetwork H, i.e., $\tau_{H^\complement} = \infty$. The events $[\gamma \subseteq G_k^F]$ are nested and increasing, so the limit is the union, and (3.26) follows. Note that G_k^F is recurrent, so $\gamma \subseteq G_k^F$ implies $\tau_x < \infty$. □

Remark 3.19. As indicated, the argument outlined above is incomplete due to the second equality of (3.27). While the set of paths from y to x in G_k^F is the same as the set of paths from y to x in G which lie in G_k, the probability of a given path may differ when computed in network or the other. This happens precisely when γ passes through a boundary point: the transition probability away from a point in bd G_k is strictly larger in G_k^F than it is in G_k.

2 Proof intuition

An irreducible walk cannot be trapped forever in a finite portion of an infinite connected graph. From each vertex of a finite set there is a positive-probability finite path to the outside. Finiteness lets us choose one common time bound and one common positive lower bound for the chance of escape. Repeating the estimate in time blocks makes eternal survival have probability zero. The event in (1) is a countable union of such null events, so ordinary conditioning on it is undefined.

3 The null-event theorem

Let $(X_n)_{n\geq 0}$ be the network random walk and, for a vertex set F, write $\tau_{F^c} := \inf\{n \geq 0 : X_n \notin F\}$.

Theorem 1. *Let (G, c) be an infinite connected resistance network in the sense of [1]. For every starting vertex y and every finite vertex set F,*

$$\mathbb{P}_y(X_n \in F \text{ for every } n \geq 0) = 0.$$

Consequently,

$$\mathbb{P}_y(|\gamma| < \infty) = 0.$$

Proof. If $y \notin F$, the first assertion is immediate, so suppose $y \in F$. For every $z \in F$, connectedness and infinitude give a finite path

$$z = z_0^{(z)}, z_1^{(z)}, \ldots, z_{\ell_z}^{(z)}$$

whose last vertex is outside F and whose earlier vertices lie in F. Every edge on this path has positive conductance and hence positive transition probability. Put

$$L := \max_{z\in F} \ell_z, \qquad q := \min_{z\in F} \prod_{j=0}^{\ell_z - 1} p\big(z_j^{(z)}, z_{j+1}^{(z)}\big).$$

The minima and maxima are over a finite set, so $L < \infty$ and $q > 0$. Starting from any $z \in F$, following the selected path shows

$$\mathbb{P}_z(\tau_{F^c} \leq L) \geq q.$$

Applying the Markov property at the deterministic times $L, 2L, \ldots$ therefore yields, for every $m \geq 1$,

$$\mathbb{P}_y(\tau_{F^c} > mL) \leq (1 - q)^m.$$

Letting $m \to \infty$ proves that the walk cannot stay in F forever.

The paper fixes an increasing exhaustion $G = \bigcup_{k\geq 1} G_k$ by finite subgraphs. A trajectory has finite range if and only if its range is contained in some G_k. Thus

$$\{|\gamma| < \infty\} = \bigcup_{k\geq 1} \{X_n \in G_k \text{ for every } n \geq 0\}.$$

Every event on the right is null by the first part, and the union is countable. This proves the second assertion. □

Corollary 1. *Under the usual definition $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ for $\mathbb{P}(B) > 0$, the right-hand side of (1) is undefined on every infinite resistance network. Therefore Conjecture 3.18 is not a valid probabilistic identity as stated. The same problem affects every conditional probability $\mathbb{P}_y[\tau_x < \tau_o \mid \gamma \subseteq G_k^{\mathrm{F}}]$ in the proposed equation (3.27).*

Proof. Theorem 1 gives zero probability to the conditioning event in (1). It also gives zero probability to eternal containment in the finite vertex set of each G_k^{F}. Both conditional probabilities are therefore 0/0 expressions under ordinary conditioning. □

4 What a correct replacement would require

The finite free-network law $\mathbb{P}_y^{G_k^{\mathrm{F}}}$ is a different Markov law, not a conditional version of $\mathbb{P}_y$: deleting outward edges changes a boundary transition from $p(z,w) = c_{zw}/c(z)$ to

$$p_k(z,w) = \frac{c_{zw}}{\sum_{u \in G_k} c_{zu}}.$$

This is exactly the boundary discrepancy noted in Remark 3.19 of the source. One may define a meaningful substitute using a specified limit of the laws $\mathbb{P}_y^{G_k^{\mathrm{F}}}$, a finite-time survival conditioning, a Doob transform, or another zero-event regularization. Such choices need not be equivalent, and none is specified in Conjecture 3.18. The null-event theorem therefore resolves the statement as written while leaving open the design and analysis of a revised probabilistic representation.

Weihrauch and Bachmann later proved that a different simple hitting-probability representation attributed to Corollaries 3.13 and 3.15 of [1] fails in general, and characterized its validity for transient locally finite graphs [2]. Their result reinforces the need to distinguish finite-network laws from the original infinite-network law, but it does not define or repair the null conditioning in Conjecture 3.18.

Verification, novelty, and scope

The proof uses only connectedness, infinitude, positive transition probability along edges, and a countable finite exhaustion; it does not require local finiteness, transience, or recurrence. The uniform escape constant is valid even when vertices have infinite degree because only one finite path per vertex of F is selected.

The novelty search was bounded to the run's lightweight indexes and web/arXiv searches for the source identifier, exact title, "bounded trajectory", "finite subnetwork", "probabilistic representation", and citations to the source paper. It found [2], which explicitly treats a different formula, but no later source explicitly addressing Conjecture 3.18's null event. Novelty is therefore provisional. Human review should verify the source interpretation and the finite-set escape estimate, and should keep the result scoped to ordinary conditioning; a separately specified limiting conditioning would constitute a revised problem.

References

[1] P. E. T. Jorgensen and E. P. J. Pearse, *A Hilbert space approach to effective resistance metric*, Complex Analysis and Operator Theory 4 (2010), 975–1013; arXiv:0906.2535v3.

[2] T. Weihrauch and S. Bachmann, *On the Probabilistic Representation of the Free Effective Resistance of Infinite Graphs*, Journal of Theoretical Probability 36 (2023), 1956–1971; arXiv:1902.01110.